



2 Real Analysis 2023

Tutorial 8

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The ratio and the root tests

Problem 29 For each of the following series, determine whether it converges or diverges.

$$\begin{array}{lll}
 \text{(a)} \sum_{n=1}^{\infty} \frac{n!}{n^n}, & \text{(b)} \sum_{n=1}^{\infty} \frac{2^n n!}{n^n}, & \text{(c)} \sum_{n=1}^{\infty} \frac{3^n n!}{n^n}, \\
 \text{(d)} \sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}, & \text{(e)} \sum_{n=1}^{\infty} \frac{5^n (n!)^2}{(2n)!}, & \text{(f)} \sum_{n=1}^{\infty} \left(\frac{n}{n+1} \right)^{n^2}, \\
 \text{(g)} \sum_{n=1}^{\infty} 4^n \left(\frac{n}{n+1} \right)^{n^2}, & \text{(h)} \sum_{n=1}^{\infty} \frac{n^4}{2^n}, & \text{(i)} \sum_{n=1}^{\infty} \frac{(2n)!}{n! n^n}.
 \end{array}$$

Solution.

Notice that all series in this problem have positive terms.

(a) We apply the Ratio Test:

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1)!}{(n+1)^{n+1}} \frac{n^n}{n!} = \lim_{n \rightarrow \infty} \frac{n^n}{(n+1)^n} = \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n}\right)^n} = \frac{1}{e} < 1.$$

Thus, the series $\sum_{n=1}^{\infty} \frac{n!}{n^n}$ converges.

(b) Similarly,

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{2^{n+1}(n+1)!}{(n+1)^{n+1}} \frac{n^n}{2^n n!} = \lim_{n \rightarrow \infty} \frac{2}{\left(1 + \frac{1}{n}\right)^n} = \frac{2}{e} < 1.$$

Thus, the series $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$ converges.

(c) Following the same lines, we obtain

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{3^{n+1}(n+1)!}{(n+1)^{n+1}} \frac{n^n}{3^n n!} = \lim_{n \rightarrow \infty} \frac{3}{\left(1 + \frac{1}{n}\right)^n} = \frac{3}{e} > 1.$$

Thus, the series $\sum_{n=1}^{\infty} \frac{3^n n!}{n^n}$ diverges.

(d) We apply the Ratio Test:

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1)!(n+1)!}{(2n+2)!} \frac{(2n)!}{n! n!} = \lim_{n \rightarrow \infty} \frac{(n+1)(n+1)}{(2n+2)(2n+1)}$$

$$= \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{1}{n}\right) \left(1 + \frac{1}{n}\right)}{\left(2 + \frac{2}{n}\right) \left(2 + \frac{1}{n}\right)} = \frac{1}{4} < 1.$$

Thus, the series $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$ converges.

(e) Similarly,

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{5^{n+1}(n+1)!(n+1)!}{(2n+2)!} \frac{(2n)!}{5^n n! n!} = \lim_{n \rightarrow \infty} \frac{5(n+1)(n+1)}{(2n+2)(2n+1)} = \frac{5}{4} > 1.$$

Thus, the series $\sum_{n=1}^{\infty} \frac{5^n (n!)^2}{(2n)!}$ diverges.

(f) We apply the Root Test:

$$L = \lim_{n \rightarrow \infty} a_n^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left\{ \left(\frac{n}{n+1} \right)^{n^2} \right\}^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n = \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n}\right)^n} = \frac{1}{e} < 1.$$

Thus, the series $\sum_{n=1}^{\infty} \left(\frac{n}{n+1} \right)^{n^2}$ converges.

(g) Similarly,

$$L = \lim_{n \rightarrow \infty} a_n^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left\{ 4^n \left(\frac{n}{n+1} \right)^{n^2} \right\}^{\frac{1}{n}} = \lim_{n \rightarrow \infty} 4 \left(\frac{n}{n+1} \right)^n = \lim_{n \rightarrow \infty} \frac{4}{\left(1 + \frac{1}{n}\right)^n} = \frac{4}{e} > 1.$$

Thus, the series $\sum_{n=1}^{\infty} 4^n \left(\frac{n}{n+1} \right)^{n^2}$ diverges.

(h) We apply the Ratio Test:

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^4 2^n}{2^{n+1} n^4} = \lim_{n \rightarrow \infty} \frac{1}{2} \left(1 + \frac{1}{n}\right)^4 = \frac{1}{2} < 1.$$

Thus, the series $\sum_{n=1}^{\infty} \frac{n^4}{2^n}$ converges.

(i) We apply the Ratio Test:

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(2n+2)!}{(n+1)!(n+1)^{n+1}} \frac{n! n^n}{(2n)!} = \lim_{n \rightarrow \infty} \frac{(2n+2)(2n+1)}{(n+1)(n+1)} \left(\frac{n}{n+1} \right)^n \\ &= \lim_{n \rightarrow \infty} \frac{\left(2 + \frac{2}{n}\right) \left(2 + \frac{1}{n}\right)}{\left(1 + \frac{1}{n}\right) \left(1 + \frac{1}{n}\right)} \frac{1}{\left(1 + \frac{1}{n}\right)^n} = \frac{4}{e} > 1. \end{aligned}$$

Thus, the series $\sum_{n=1}^{\infty} \frac{(2n)!}{n! n^n}$ diverges.

Absolute and conditional convergence

Problem 30 For a real number x , we define $x^+ = \max\{0, x\}$ and $x^- = \max\{0, -x\}$.

(a) Show that

$$x = x^+ - x^-, \quad |x| = x^+ + x^-, \quad x^+ = \frac{1}{2}(|x| + x), \quad x^- = \frac{1}{2}(|x| - x).$$

(b) Let $\sum_{n=1}^{\infty} a_n$ be a real series. Show that $\sum_{n=1}^{\infty} a_n$ is absolutely convergent if and only if both series $\sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} a_n^-$ are convergent.

(c) Show that if one of the series $\sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} a_n^-$ converges and the other diverges, then $\sum_{n=1}^{\infty} a_n$ diverges.

(d) Show that if $\sum_{n=1}^{\infty} a_n$ is conditionally convergent, then both series $\sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} a_n^-$ diverge.

Solution.

(a) If $x \geq 0$, then $x^+ = x$, $x^- = 0$, so that $x = x^+ - x^-$ and $|x| = x = x^+ + x^-$.

If $x < 0$, then $x^+ = 0$, $x^- = -x$, so that $x = -x^- = x^+ - x^-$ and $|x| = -x = x^- = x^+ + x^-$.

If we add the two equations $x = x^+ - x^-$ and $|x| = x^+ + x^-$, we obtain that $x + |x| = 2x^+$, and this implies $x^+ = \frac{1}{2}(|x| + x)$.

If we subtract the equation $x = x^+ - x^-$ from $|x| = x^+ + x^-$, we obtain that $|x| - x = 2x^-$, and this implies $x^- = \frac{1}{2}(|x| - x)$.

(b) First suppose that both series $\sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} a_n^-$ are convergent. Then for any $N \in \mathbb{N}$

$$\sum_{n=1}^N |a_n| = \sum_{n=1}^N a_n^+ + \sum_{n=1}^N a_n^-$$

and the limit

$$\lim_{N \rightarrow \infty} \sum_{n=1}^N |a_n| = \lim_{N \rightarrow \infty} \sum_{n=1}^N a_n^+ + \lim_{N \rightarrow \infty} \sum_{n=1}^N a_n^-$$

exists. It follows that $\sum_{n=1}^{\infty} |a_n|$ converges, and hence $\sum_{n=1}^{\infty} a_n$ converges absolutely.

Now suppose that $\sum_{n=1}^{\infty} a_n$ converges absolutely. By the definition of absolute convergence, $\sum_{n=1}^{\infty} |a_n|$ converges. For any $N \in \mathbb{N}$ we have

$$\sum_{n=1}^N a_n^+ = \frac{1}{2} \left(\sum_{n=1}^N |a_n| + \sum_{n=1}^N a_n \right), \quad \sum_{n=1}^N a_n^- = \frac{1}{2} \left(\sum_{n=1}^N |a_n| - \sum_{n=1}^N a_n \right),$$

and the limits $\lim_{N \rightarrow \infty} \sum_{n=1}^N a_n^+$, $\lim_{N \rightarrow \infty} \sum_{n=1}^N a_n^-$ exist by the Algebraic Limit Theorem. It follows that the series $\sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} a_n^-$ converge.

- (c) Assume without loss of generality that $\sum_{n=1}^{\infty} a_n^+$ converges, $\sum_{n=1}^{\infty} a_n^-$ diverges. Suppose, by contradiction, that $\sum_{n=1}^{\infty} a_n$ converges. Then for any $N \in \mathbb{N}$

$$\sum_{n=1}^N a_n^- = \sum_{n=1}^N a_n^+ - \sum_{n=1}^N a_n,$$

and the right-hand side converges as $N \rightarrow \infty$. But this implies that also the left-hand side converges, i.e., the series $\sum_{n=1}^{\infty} a_n^-$ converges. The contradiction shows that $\sum_{n=1}^{\infty} a_n$ must diverge.

- (d) This follows from (b) and (c). If both series $\sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} a_n^-$ converge, then $\sum_{n=1}^{\infty} a_n$ converges absolutely by (b). On the other hand, if one of the series $\sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} a_n^-$ converges and the other diverges, then $\sum_{n=1}^{\infty} a_n$ diverges by (c). The only possibility in the case when $\sum_{n=1}^{\infty} a_n$ converges conditionally is therefore that both series $\sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} a_n^-$ diverge.

Problem 31 Prove or disprove.

- (a) The series $\sum_{n=0}^{\infty} \left(-\frac{2}{3}\right)^n$ can be rearranged so that the sum of the rearranged series is 10.
 (b) The series $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$ can be rearranged so that the sum of the rearranged series is 10.

Solution.

- (a) The statement is false. The series $\sum_{n=0}^{\infty} \left(-\frac{2}{3}\right)^n$ converges absolutely, since $\sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n$ is a convergent geometric series. It follows that any rearrangement of $\sum_{n=0}^{\infty} \left(-\frac{2}{3}\right)^n$ converges to

$$\sum_{n=0}^{\infty} \left(-\frac{2}{3}\right)^n = \frac{1}{1 - \left(-\frac{2}{3}\right)} = \frac{3}{5}.$$

- (b) The statement is true because $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$ converges conditionally, so it can be rearranged to any sum.

To see that $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$ converges conditionally, notice that the series converges by the Alternating Series Test. Indeed, for $a_n = \frac{1}{\sqrt{n+1}}$ we have $\lim_{n \rightarrow \infty} a_n = 0$ and $a_n \geq a_{n+1}$. On the other hand, the series $\sum_{n=0}^{\infty} \left| \frac{(-1)^n}{\sqrt{n+1}} \right| = \sum_{n=0}^{\infty} \frac{1}{\sqrt{n+1}}$ is a divergent p -series with $p = \frac{1}{2}$.

Problem 32 For each of the following series, determine whether it is absolutely convergent, conditionally convergent or divergent.

- | | | |
|---|---|---|
| (a) $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n^2 + 4},$ | (b) $\sum_{n=0}^{\infty} (-1)^n e^{-n},$ | (c) $\sum_{n=1}^{\infty} \frac{\sin \sqrt{n}}{\sqrt{n^3 + 2}},$ |
| (d) $\sum_{n=1}^{\infty} (-1)^n \frac{\arctan n}{n^2},$ | (e) $\sum_{n=1}^{\infty} (-1)^n \frac{n^4}{e^n},$ | (f) $\sum_{n=1}^{\infty} \left(\frac{-n}{2n+1} \right)^n.$ |

Solution.

- (a) The series $\sum_{n=1}^{\infty} \frac{n}{n^2+4}$ diverges by the Comparison Test: we have for $n \geq 2$

$$\frac{n}{n^2+4} \geq \frac{n}{n^2+n^2} = \frac{1}{2n},$$

and the series $\sum_{n=1}^{\infty} \frac{1}{2n}$ diverges. Thus, $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n^2+4}$ does not converge absolutely.

The series $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n^2+4}$ converges by the Alternating Series Test. Indeed, this is an alternating series, $\lim_{n \rightarrow \infty} \frac{n}{n^2+4} = 0$ and $\frac{n+1}{(n+1)^2+4} \leq \frac{n}{n^2+4}$ for all $n \geq 2$. There are several ways to prove the latter inequality. For example, one can consider the function $\varphi(x) = \frac{x}{x^2+4}$ and show that its derivative is negative for $x > 2$. Alternatively, the inequality $\frac{n+1}{(n+1)^2+4} \leq \frac{n}{n^2+4}$ is equivalent to $(n+1)(n^2+4) \leq n((n+1)^2+4)$. The latter can be simplified to $4 \leq n^2 + n$, and this is true for $n \geq 2$.

We conclude that the series $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n^2+4}$ converges conditionally.

- (b) $\sum_{n=0}^{\infty} (-1)^n e^{-n} = \sum_{n=0}^{\infty} \left(-\frac{1}{e}\right)^n$ is a geometric series and it is absolutely convergent.

- (c) We have for $n \in \mathbb{N}$

$$\left| \frac{\sin \sqrt{n}}{\sqrt{n^3+2}} \right| \leq \frac{1}{\sqrt{n^3+2}} \leq \frac{1}{n^{3/2}},$$

and the p -series $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ converges. Thus, by the Comparison Test the series $\sum_{n=1}^{\infty} \frac{\sin \sqrt{n}}{\sqrt{n^3+2}}$ converges absolutely.

- (d) For $n \in \mathbb{N}$ we have $0 < \arctan n \leq \frac{\pi}{2}$, and therefore

$$0 \leq \frac{\arctan n}{n^2} \leq \frac{\pi}{2} \frac{1}{n^2}.$$

The series $\sum_{n=1}^{\infty} \frac{\pi}{2} \frac{1}{n^2}$ converges. By the Comparison Test, $\sum_{n=1}^{\infty} (-1)^n \frac{\arctan n}{n^2}$ converges absolutely.

- (e) We apply the Ratio Test:

$$L = \lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|} = \lim_{n \rightarrow \infty} \frac{(n+1)^2 e^n}{e^{n+1} n^4} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^4 \frac{1}{e} = \frac{1}{e} < 1.$$

We conclude $\sum_{n=1}^{\infty} (-1)^n \frac{n^4}{e^n}$ converges absolutely.

- (f) We apply the Root Test:

$$L = \lim_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{n}{2n+1} = \frac{1}{2}.$$

We conclude $\sum_{n=1}^{\infty} \left(\frac{-n}{2n+1}\right)^n$ converges absolutely.

Mixed problems on series

Problem 33 For each of the following series, determine whether it converges or diverges.

$$\begin{array}{lll} \text{(a)} \sum_{n=1}^{\infty} \frac{\sqrt{n+1} - \sqrt{n}}{n}, & \text{(b)} \sum_{n=1}^{\infty} \frac{1}{n} \left(\frac{2}{3}\right)^n, & \text{(c)} \sum_{n=1}^{\infty} n \left(\frac{2}{3}\right)^n, \\ \text{(d)} \sum_{n=1}^{\infty} \left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right), & \text{(e)} \sum_{n=1}^{\infty} \frac{1}{2+3\ln n}, & \text{(f)} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2+3\ln n}. \end{array}$$

Solution.

(a) For $n \in \mathbb{N}$

$$\begin{aligned} 0 < \frac{\sqrt{n+1} - \sqrt{n}}{n} &= \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{n(\sqrt{n+1} + \sqrt{n})} = \frac{1}{n(\sqrt{n+1} + \sqrt{n})} \\ &\leq \frac{1}{n(\sqrt{n} + \sqrt{n})} = \frac{1}{2n^{3/2}}, \end{aligned}$$

and the series $\sum_{n=1}^{\infty} \frac{1}{2n^{3/2}}$ is a convergent p -series. By the Comparison Test, $\sum_{n=1}^{\infty} \frac{\sqrt{n+1} - \sqrt{n}}{n}$ converges.

(b) For $n \in \mathbb{N}$ we have $\frac{1}{n} \left(\frac{2}{3}\right)^n \leq \left(\frac{2}{3}\right)^n$, and $\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n$ is a converging geometric series. By the Comparison Test, $\sum_{n=1}^{\infty} \frac{1}{n} \left(\frac{2}{3}\right)^n$ converges.

(c) We apply the Ratio Test:

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1) \left(\frac{2}{3}\right)^{n+1}}{n \left(\frac{2}{3}\right)^n} = \lim_{n \rightarrow \infty} \frac{n+1}{n} \cdot \frac{2}{3} = \frac{2}{3} < 1,$$

the series $\sum_{n=1}^{\infty} n \left(\frac{2}{3}\right)^n$ converges.

(d) This is a telescopic series. For its partial sums we have

$$\begin{aligned} s_N &= \sum_{n=1}^N \left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right) = 1 - \frac{1}{2^2} + \frac{1}{2^2} - \frac{1}{3^2} + \cdots + \frac{1}{N^2} - \frac{1}{(N+1)^2} \\ &= 1 - \frac{1}{(N+1)^2} \rightarrow 1, \quad N \rightarrow \infty. \end{aligned}$$

The series $\sum_{n=1}^{\infty} \left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right)$ converges.

(e) For $n \geq 2$ we have

$$\frac{1}{2+3\ln n} \geq \frac{1}{2+3n} \geq \frac{1}{4n},$$

and the series $\sum_{n=1}^{\infty} \frac{1}{4n}$ diverges. By the Comparison Test, $\sum_{n=1}^{\infty} \frac{1}{2+3\ln n}$ diverges.

(f) The series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2+3\ln n}$ is an alternating series, $\lim_{n \rightarrow \infty} \frac{1}{2+3\ln n} = 0$ and $\frac{1}{2+3\ln(n+1)} \leq \frac{1}{2+3\ln n}$ (because $\ln(n+1) \geq \ln n$). By the Alternating Series Test, it converges.